



NCLEX IMMUNODEFICIENCY OVERVIEW

1. WHAT IS IMMUNODEFICIENCY?

- Immunodeficiency is a condition where the immune system is weak or not functioning properly, leading to increased risk of infections, cancers, and autoimmune diseases.
- Two Types:
 - ✓ Primary Immunodeficiency (Born with a weak immune system – Genetic or Congenital).
 - ✓ Secondary Immunodeficiency (Caused by another condition – Acquired later in life).

 **NCLEX Tip:** Primary = Genetic (From Birth), Secondary = Acquired (Due to Disease or Medications).

2. PRIMARY IMMUNODEFICIENCY DISORDERS (GENETIC/INHERITED)

Condition	Cause	Symptoms	Treatment
Severe Combined Immunodeficiency (SCID)	Genetic defect → No B or T cells (White blood cells that fight infections).	Frequent severe infections (Pneumonia, ear infections), failure to thrive.	Bone marrow transplant (Only cure), IV immunoglobulin (IVIG) therapy.
X-Linked Agammaglobulinemia (Bruton's Disease)	No B cells → No antibodies.	Recurrent bacterial infections.	IVIG (Provides antibodies).
Common Variable Immunodeficiency (CVID)	Low antibodies, unknown cause.	Frequent respiratory infections.	IVIG.
DiGeorge Syndrome (Thymic Hypoplasia)	Missing thymus → No T cells.	Heart defects, cleft palate,	Thymus transplant.

Condition	Cause	Symptoms	Treatment
		frequent infections.	

 **NCLEX Tip:** Primary immunodeficiencies often require lifelong IV immunoglobulin (IVIG) therapy!

3. SECONDARY IMMUNODEFICIENCY DISORDERS (ACQUIRED LATER IN LIFE)

♦ Caused by:

- ✓ HIV/AIDS (Destroys CD4 immune cells).
- ✓ Cancer (Leukemia, Lymphoma – Attack immune cells).
- ✓ Chemotherapy & Radiation (Kills fast-growing immune cells).
- ✓ Steroids (Suppress immune response – Used in autoimmune diseases).
- ✓ Malnutrition (Lack of vitamins weakens the immune system).
- ✓ Diabetes (High blood sugar weakens immune response).

 **NCLEX Tip:** Steroids (Prednisone) & Chemotherapy = Major Causes of Immunosuppression!

4. SYMPTOMS OF IMMUNODEFICIENCY

♦ Mild to Moderate Symptoms:

- ✗ Frequent colds, sinus infections, ear infections.
- ✗ Slow healing wounds.
- ✗ Fatigue, weight loss.

♦ Severe Symptoms (Life-Threatening Infections):

- ✗ Pneumonia, Tuberculosis (Lung Infections).
- ✗ Opportunistic Infections (Infections that rarely happen in healthy people, e.g., Pneumocystis pneumonia in HIV/AIDS).
- ✗ Chronic diarrhea, fungal infections (Oral Thrush, Yeast Infections).

 **NCLEX Tip: Opportunistic infections are the biggest concern in immunodeficiency!**

5. DIAGNOSIS OF IMMUNODEFICIENCY

Test	What It Detects
Complete Blood Count (CBC) with Differential	Looks at WBC count (Low WBC = Immunodeficiency).
Immunoglobulin Levels (IgA, IgG, IgM, IgE)	Checks antibody levels to assess immune function.
T-Cell & B-Cell Count	Measures specific immune cells (Low = Immunodeficiency).
HIV Test (ELISA, Western Blot, RNA Test)	Checks for HIV infection (A cause of secondary immunodeficiency).

 **NCLEX Tip: Low WBC, Low Immunoglobulins = Immunodeficiency!**

6. TREATMENT OF IMMUNODEFICIENCY

- ✓ **IV Immunoglobulin (IVIG) Therapy (For Antibody Deficiencies – Provides Passive Immunity).**
- ✓ **Bone Marrow Transplant (For Severe Genetic Conditions like SCID).**
- ✓ **Antibiotic Prophylaxis (Prevent Infections Before They Happen).**
- ✓ **Antiretroviral Therapy (ART) for HIV/AIDS (Prevents Immune Decline).**
- ✓ **Avoid Live Vaccines (Weakened Immune Systems Can't Handle Them).**

 **NCLEX Tip: No Live Vaccines for Immunocompromised Patients (MMR, Varicella, Flu Mist)!**

7. PREVENTING INFECTIONS IN IMMUNODEFICIENCY PATIENTS

- ✓ Hand Hygiene – #1 Way to Prevent Infections!
- ✓ Avoid Sick Contacts & Crowded Places.
- ✓ Cook Meat & Eggs Fully (Avoid Foodborne Illnesses).
- ✓ No Gardening (Risk of Fungal Infections in Soil – Aspergillosis).
- ✓ Use Masks in Public (Reduces Exposure to Airborne Germs).

🚨 NCLEX Tip: Immunocompromised patients should NOT clean litter boxes (Risk of Toxoplasmosis - Parasite Infection)!

8. COMPLICATIONS OF IMMUNODEFICIENCY

- ⚠️ Frequent & Severe Infections (Leading Cause of Death).
- ⚠️ Opportunistic Infections (Fungal, Viral, & Bacterial Infections).
- ⚠️ Cancers (Higher Risk of Lymphoma, Skin Cancer).
- ⚠️ Autoimmune Diseases (Body Attacks Itself).

🚨 NCLEX Tip: Patients with immunodeficiency have a higher risk of developing autoimmune diseases like lupus & rheumatoid arthritis!

9. NCLEX NURSING CARE FOR IMMUNODEFICIENCY

- ✓ Monitor for Signs of Infection (Fever, Cough, Wounds Not Healing).
- ✓ Encourage Vaccinations (But No Live Vaccines!).
- ✓ Educate on Infection Prevention (Handwashing, Avoiding Crowds).
- ✓ Ensure Adequate Nutrition (Protein & Vitamins Boost Immune Function).
- ✓ Teach Patients to Take Medications Exactly as Prescribed (ART for HIV, IVIG for Primary Deficiencies).

10. NCLEX "MUST-KNOW"

IMMUNODEFICIENCY TIPS

- ♦ **Primary = Genetic (From Birth), Secondary = Acquired (Due to Disease/ Drugs).**
- ♦ **SCID = No B or T Cells → Bone Marrow Transplant Needed!**
- ♦ **HIV/AIDS = Most Common Cause of Secondary Immunodeficiency.**
- ♦ **Opportunistic Infections = Major Risk (Pneumocystis Pneumonia, Oral Thrush, TB).**
- ♦ **No Live Vaccines for Immunocompromised Patients!**
- ♦ **IVIG Provides Passive Immunity in Antibody Deficiencies.**
- ♦ **Prevent Infection → Hand Hygiene, Avoid Raw Foods, No Cat Litter!**